

Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

November 7, 2019

Day 3: Random Variables: Discrete

- Random Variables
- Expectation, Variance
- Discrete Probability Mass Function
- Cumulative Distribution Functions
- Joint Probability Mass Functions

Random Variables

Given a sample space Ω , and a probability law P :

Definition

*A random variable is a **function** that assigns real numbers (usually) to events in a sample space Ω .*

$$X(\omega) : \Omega \rightarrow \mathbb{R}$$

Random Variables: Example

Sample space from tossing three coins:

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

Let $Y(\omega)$ be the number of heads in outcome $\omega \in \Omega$.

- $Y(HHH) = 3$
- $Y(HHT) = Y(HTH) = Y(THH) = 2$
- $Y(HTT) = Y(THT) = Y(TTH) = 1$
- $Y(TTT) = 0$

Random Variables: Example II

Sample space from the number of dots on a die roll.

$$\Omega = \left\{ \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array}, \begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array} \right\}$$

Let $Z(\omega)$ be the dots on the die face $\omega \in \Omega$.

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=1$$

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=2$$

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=3$$

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=4$$

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=5$$

$$\blacksquare Z(\begin{array}{|c|c|c|c|c|c|} \hline \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline & & & & & \\ \hline \end{array})=6$$

Random Variables: Example III

Sample space from the number of dots on a die roll.

$$\Omega = \{ \blacksquare, \blacksquare\bullet, \blacksquare\bullet\bullet, \blacksquare\bullet\bullet\bullet, \blacksquare\bullet\bullet\bullet\bullet, \blacksquare\bullet\bullet\bullet\bullet\bullet \}$$

Let $X(\omega)$ be 3 to the power of the number of dots on the die face $\omega \in \Omega$ minus 1.

$$\blacksquare X(\blacksquare) = 3^{1-1} = 3^0 = 1$$

$$\blacksquare X(\blacksquare\bullet) = 3^{2-1} = 3$$

$$\blacksquare X(\blacksquare\bullet\bullet) = 3^{3-1} = 9$$

$$\blacksquare X(\blacksquare\bullet\bullet\bullet) = 3^{4-1} = 27$$

$$\blacksquare X(\blacksquare\bullet\bullet\bullet\bullet) = 3^{5-1} = 81$$

$$\blacksquare X(\blacksquare\bullet\bullet\bullet\bullet\bullet) = 3^{6-1} = 243$$

$$X(\omega) = 3^{\omega-1}$$

functions

- Random Variables are functions.
- In R, we see functions look like a word followed by ():
 - mean(), median(), head(), str()
- Our own functions are written using:

```
function(input){  
  do something  
  return(output)}
```

Properties of Random Variables

Random variables (for us) will always give a number.

- X, Y, Z : capital letters for a random variable.
- x, y, z : are the output of the function.

Kinds of Random Variables

Definition

A random variable may be

- 1** *discrete: if outcomes are countable.*
- 2** *continuous: If outcomes can be any value in some interval.*
- 3** *mixed: if outcomes are a combination of events and intervals.*

Connecting random variables to probability

- X assigns some numbers to events.
- Remember, probability assigns a likelihood to an event.
- What is the probability that $X(\omega)$ gives some number x ?
- $P(X = x)$
- This is another function (of a function)

Discrete Probability Function

Definition

If X is a discrete random variable, the probability of each outcome x is provided by a probability function:

$$p_X(x) = P(X = x) = P(\{\omega \in \Omega : X(\omega) = x\})$$

- this is also called the *probability mass function* (pmf)
- We often just write $p(x)$, some people use $f_X(x)$

Building a Discrete Probability Function: Example

- Sample space: $\Omega = \{\text{Pashto}, \text{Not Pashto}\}$
- $X=1$ if $\omega = \text{Pashto}$
- $X=0$ if $\omega = \text{Not Pashto}$
- $P(X=1)=.3$

A pmf can take on any value of x , so we must consider all possibilities of x .

- $x = 1$
- $x = 0$
- all other values of x .

- If $x = 1$

$$P(X = x) = P(X = 1) = P(\text{Pashto}) = p = .3$$

- If $x = 0$

$$P(X = x) = P(X = 0) = P(\text{Not Pashto}) = 1 - p = .7$$

- all other values of x .

$$P(X = x) = 0$$

Combining these results, we get the probability function.

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

This can be written shorthand as:

$$f_X(x) = p^x(1 - p)^{(1-x)}, \quad x \in \{0, 1\}$$

Where the 0 is left implicit.

Dichotomous process

To summarize

- If X only has two values 0 and 1 and $p = P(X = 1)$ then X has the probability function:

$$f_X(x) = p^x(1 - p)^{(1-x)}, \quad x \in \{0, 1\}$$

- This is called a Bernoulli distribution:

$$X \sim \text{Bernoulli}(p)$$

- p is a *parameter*

Bernoulli Random Variable

Suppose we flip a fair coin and $Y = 1$ if the outcome is Heads .

$$Y \sim \text{Bernoulli}(1/2)$$

$$p(1) = (1/2)^1(1 - 1/2)^{1-1} = 1/2$$

$$p(0) = (1/2)^0(1 - 1/2)^{1-0} = (1 - 1/2)$$

Example: Winning a War

Suppose country 1 is engaged in a conflict and can either win or lose. Define $Y = 1$ if the country wins and $Y = 0$ otherwise.

Then,

$$Y \sim \text{Bernoulli}(\pi)$$

Suppose country 1 is deciding whether to fight a war.

Engaging in the war will cost the country c .

If they win, country 1 receives B .

Example: Sequence of Bernoulli trials

We might be interested in the outcome of several independent trials **ie. with replacement**.

Consider a coin, flipped twice. Suppose $X_1(T) = 0$, $X_1(H) = 1$,
 $X_2(T) = 0$, $X_2(H) = 1$

$$P(X_1 = 1, X_2 = 1) = P(X_1 = 1 \cap X_2 = 1) = P(X_1 = 1)P(X_2 = 1)$$

$$p(x_1, x_2) = p^{x_1}(1-p)^{1-x_1}p^{x_2}(1-p)^{1-x_2}$$

$$p(x_1, x_2) = p^{x_1+x_2}(1-p)^{2-x_1-x_2}$$

Let $X_1, \dots, X_n$ be independent Bernoulli random variables with the same parameter p .

$$p(x_1, \dots, x_n) = p(x_1)p(x_2)\dots p(x_n) = p^{x_1+\dots+x_n}(1-p)^{n-x_1-\dots-x_n}$$

$$\text{for } x_i \in \{0, 1\} \text{ and } i = 1, \dots, n$$

Expectation

What can we **expect** from a trial?

Expectation informally

The expectation is based on the principle that the value of a future gain should be directly proportional to the chance of getting it.

$$E[X + Y] = E[X] + E[Y]$$

$$E[a] = a$$

$$E[aX] = aE[X]$$

$$E[E[X]] = E[X]$$

$$E[XY] \neq E[X] \times E[Y]$$

Properties of the Expectation

$$E[a] = a$$

Proof: Suppose Y is a random variable $= a$ with probability 1 and 0 otherwise:

$$E[Y] = \sum_{y:p(y)>0} yp(y) = a * 1$$

Also:

$$E[aX] = \sum_{x:p(x)>0} a x p(x)$$

$$E[aX] = a \sum_{x:p(x)>0} x p(x)$$

Indicator Variables and Probabilities



Variance

Expected value is a measure of **central tendency**.

Variance

Definition

The variance of a random variable X , $\text{var}(X)$, is

$$\begin{aligned}\text{var}(X) &= E[(X - E[X])^2] \\ &= E[X^2] - E[X]^2\end{aligned}$$

- We will define the standard deviation of X , $\text{sd}(X) = \sqrt{\text{var}(X)}$
- $\text{var}(X) \geq 0$.

Variance Corollary

Corollary

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Proof.



Bernoulli Random Variable Moments

Suppose $Y \sim \text{Bernoulli}(\pi)$

Trials with More than Two Outcomes

Definition

Suppose we observe a trial, which might result in J outcomes.

And that $P(\text{outcome} = i) = \pi_i$

$\mathbf{Y} = (Y_1, Y_2, \dots, Y_J)$ where $Y_j = 1$ if outcome j occurred and 0 otherwise.

*Then $\mathbf{Y}$ follows a **multinomial** distribution, with*

$$p(\mathbf{y}) = \pi_1^{y_1} \pi_2^{y_2} \dots \pi_k^{y_k}$$

if $\sum_{i=1}^k y_i = 1$ and the pmf is 0 otherwise.

Equivalently, we'll write

$$\mathbf{Y} \sim \text{Multinomial}(1, \boldsymbol{\pi})$$

$$\mathbf{Y} \sim \text{Categorical}(\boldsymbol{\pi})$$

Multinomial Properties + Notes

Computer scientists: commonly call $\text{Multinomial}(1, \boldsymbol{\pi})$ **Discrete**($\boldsymbol{\pi}$).

$$\begin{aligned} E[X_i] &= N\pi_i \\ \text{var}(X_i) &= N\pi_i(1 - \pi_i) \end{aligned}$$

Counting the Number of Events

Often interested in counting number of events that occur:

- 1) Number of wars started
- 2) Number of speeches made
- 3) Number of bribes offered
- 4) Number of people waiting for license

Generally referred to as **event counts**

Stochastic processes: a course provide introduction to many processes
(**Queing Theory**)

Definition

Suppose X is a random variable that counts the number of successes in N independent and identically distributed Bernoulli trials. Then X is a *Binomial* random variable,

$$p(k) = \binom{N}{k} \pi^k (1 - \pi)^{1-k}$$

for $k \in \{0, 1, 2, \dots, N\}$ and $p(k) = 0$ otherwise.
Equivalently,

$$Y \sim \text{Binomial}(N, \pi)$$

Binomial Random Variable Moments

$$Z = \sum_{i=1}^N Y_i \text{ where } Y_i \sim \text{Bernoulli}(\pi)$$

Example: Counts

We might be interested in the *sum* of several independent trials:

- n draws from a population size N with replacement.
- If Y is number of the sample that have a binary characteristic (support Trump, show up heads, etc).
- and M is the number of the people in the population that have a binary characteristic.

Our random variable, Y , is equal to the sum of successes. Let y be an integer between 0 and n ,

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{(n-y)}$$

where $p = \frac{M}{N}$

Binomial Random Variable

- A model to count the number of successes across N trials

Are we done?

Origins of Binomial

One head on one Coin:

$$P(Y = 1) = p$$

One head on two coins:

$$P(Y = 1) = p(1 - p) + (1 - p)p = 2 * p(1 - p)^1$$

One head on three coins:

$$P(Y = 1) = p(1 - p)(1 - p) + (1 - p)p(1 - p) + (1 - p)(1 - p)p = 3 * p(1 - p)^2$$

One head on four coins:

$$\begin{aligned} P(Y = 1) &= p(1 - p)^3 + (1 - p)p(1 - p)^2 \\ &+ (1 - p)^2 p(1 - p) + (1 - p)^3 p = 4 * p(1 - p)^3 \end{aligned}$$

Origins of Binomial

One head on four coins

$$P(Y = 1) = 4 * p(1 - p)^3$$

Two heads on four coins

$$P(Y = 2) = 6 * p^2(1 - p)^2$$

Three heads on four coins

$$P(Y = 3) = 4 * p^3(1 - p)$$

Number of different realizations with y successes: $\binom{n}{y}$

$$P(Y = y) = \binom{n}{y} p^y (1 - p)^{(n-y)}$$

where $p = \frac{M}{N}$

Origins of Binomial

If $X_1, \dots, X_n$ are Bernoulli distributed with probability p , then

$$Y = \sum_{k=1}^n X_k \sim \text{Binomial}(n, p)$$

n and p are parameters of the binomial.

Assumptions of Binomial Distribution

- The number of trials is fixed,
- The probability of success is the same for each trial.
- The trials are independent

Example: Baseball!

- The World Series of Baseball ends when one team wins 4 games.
- Each game is a Bernoulli trial.
- Length of World Series $\in \{4, 5, 6, 7\}$.
- What is the probability that the World Series lasts 7 games?
- $P(X=7)$?

Example: Counts in a time period

We might be interested in the sum of events occurring in a specific time or space.

- Number of wars in a year.
- Number of diseased villages in a district.

Let X be the number of times an event occurs in a unit of time or space.

- The probability that the event occurs in a given unit of time or space is the same for all the units.
- The number of events that occur in one unit of time is independent of the number of events in other units.

Then X is a Poisson random variable, with rate $= \lambda$ and pf:

$$p(x) = \frac{\lambda^x}{x!} e^{-\lambda} \quad x = 0, 1, 2, \dots$$

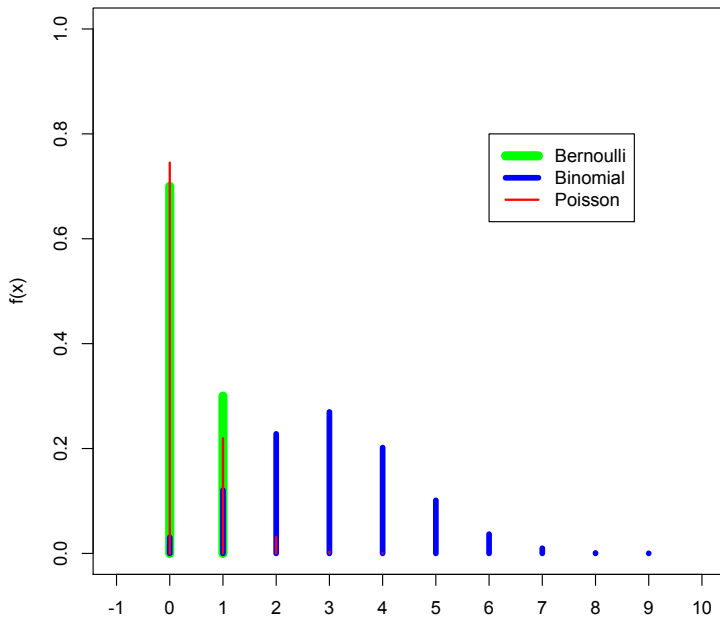
Example: Poisson Distribution

If, on average, two customers sign up every hour, what is the probability that 4 customers show up in a given hour?

$$\lambda = 2$$

$$P(X = 4) = \frac{2^4}{4!} e^{-2} = .09$$

PF



Functions of Random Variables

Definition

If $Y = g(X)$, and X is a random variable with probability mass function $p_X(x)$, Y is a random variable with pmf:

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$